25901581 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION March 2026

Fourth SEMESTER

Discipline Specific Core Course (DSC) - MG4DSCZGY203 - BIOLOGICAL BASIS OF BEHAVIOR - IV

(2024 ADMISSION ONWARDS)

Duration: 1 hour 30 minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Multiple Choice Questions

Answer all questions

Each question carries **1** marks

1. Which lymphocytes mature in the bone marrow? [K] / [CO1]

   a). T lymphocytes b). B lymphocytes

   c). Helper T cells d). Cytotoxic T cells

2. Which immunoglobulin is mainly transferred through breast milk? [K] / [CO1]

   a). IgG b). IgM

   c). IgA d). IgE

3. Explain Type IV hypersensitivity reaction? [U] / [CO2]

   a). Cell-mediated immunity b). Humoral immunity

   c). Innate immunity d). Passive immunity

4. Androgens mainly influence which function? [U] / [CO3]

   a). Ovulation b). Development of male secondary characters

c). Milk secretions d). Menstrual cycle

5. Increased muscle mass in males during puberty is mainly due to: [U] / [CO3]

a). Estrogen b). Thyroxine

c). Insulin d). Testosterone

6. Estrogen helps in regulations of [U] / [CO3]

a). Blood pressure b). Menstrual cycle

c). Digestion d). Body temperature

7. Increase in sex hormone secretion occurs mainly during [U] / [CO3]

a). Childhood b). Puberty

c). Old age d). Infancy

8. For someone who travels across time zones, what is the most immediate way to help reset their circadian rhythm? [U] / [CO4]

a). Taking melatonin upon arrival at the destination bedtime b). Seeking exposure to daylight at the appropriate local time

c). Eating according to the old timezone's schedule d). Exercising intensely upon arrival

9. Narcolepsy is a disorder marked by [U] / [CO4]

a). Delayed sleep phase b). Sudden sleep attacks

c). Loud snoring d). Reduced REM sleep

10. Which brain structure is responsible for converting neural signals from the SCN into melatonin secretion? [K] / [CO4]

a). Hypothalamus b). Pituitary gland

c). Pineal gland d). Retina

[1x10 = 10]

Part B

Short Answer Questions

Answer any **10** questions

Each question carries **2** marks

11. Write briefly about the spleen as a secondary lymphoid organ. [K] / [CO1]

12. What is rheumatoid arthritis? [K] / [CO2]

13. Mention two stress-reducing practices that improve immunity. [K] / [CO1]

14. Mention any two causes of immunodeficiency. [K] / [CO2]

15. Write a short note on the scrotum. [U] / [CO3]

16. What is the excitement phase of the sexual response cycle? [K] / [CO3]

17. State the functions of Bartholin's glands. [U] / [CO3]

18. Describe the process of development of male reproductive organs during fetal life. [U] / [CO3]

19. Describe any two symptoms of mood disorder. [U] / [CO4]

20. Describe two differences between jet lag and travel fatigue. [K] / [CO4]

21. What is infradian rhythm? [U] / [CO4]

22. What is meant by biological rhythm in living organisms? [U] / [CO4]

[2x10 = 20]

Part C

Short Essay Questions

Answer any **5** questions

Each question carries **4** marks

23. Write a short note on environmental zeitgebers. [U] / [CO4]

24. Explain the role of melatonin in sleep–wake regulation. [U] / [CO4]

25. Write on natural active immunity . [K] / [CO1]

26. What is Type II hypersensitivity and explain antibody-mediated cytotoxicity in Type II hypersensitivity. [U] / [CO2]

27. Define secondary sexual characteristics. Enumerate the secondary sexual characteristics in females. [U] / [CO3]

28. Describe female fetal sex differentiation with reference to genetic makeup and hormonal environment. [U] / [CO3]

29. Enumerate the principal features of adaptive immunity and explain their significance. [K] / [CO1]

[4x5 = 20]